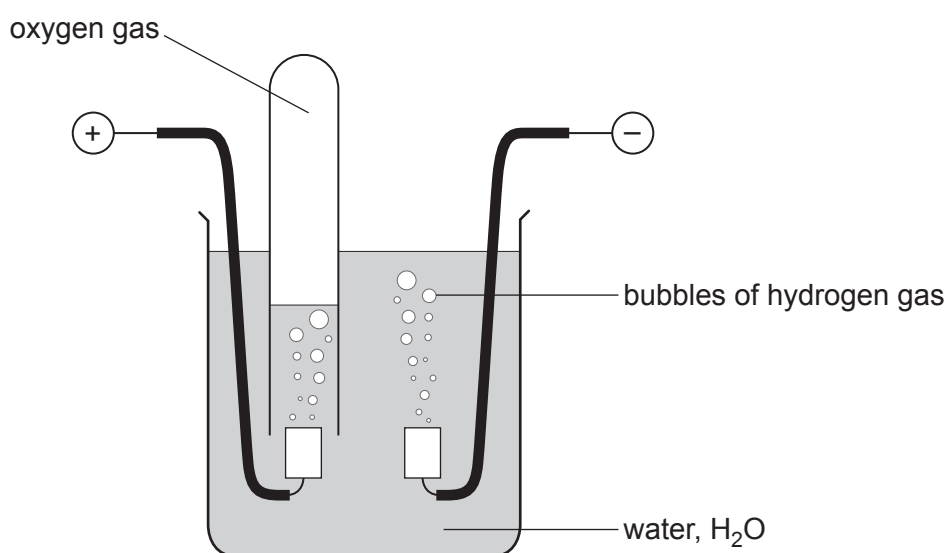


2. (a) The diagram shows the apparatus used by a student to investigate the electrolysis of water.



- (i) The student collected a sample of the oxygen formed in a test tube.

Give the test the student would carry out to show the presence of oxygen in the test tube. Include the observation the student would expect. [1]

.....

.....

- (ii) Use the formula of water, H_2O .

Give the volume of hydrogen that would form in the same time as 10 cm^3 of oxygen. [1]

..... cm^3



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(b) The table shows information about the electrolysis of three different electrolytes. The table is incomplete.

Electrolyte	Ions present in the electrolyte		Observations	
	Positive ion(s)	Negative ion(s)	At the negative (–) electrode	At the positive (+) electrode
molten lead(II) bromide	Pb ²⁺		grey metal A formed	orange gas formed
aqueous copper(II) chloride	 and H ⁺	Cl [–] and OH [–]	brown metal formed	green-yellow gas B formed
aqueous compound C	Zn ²⁺ and H ⁺	I [–] and OH [–]	grey metal formed	brown solution formed

(i) Complete the table by adding the **symbols** of the missing **ions**. [2]

(ii) Name substances **A**, **B** and **C**. [3]

Metal **A**

Gas **B**

Compound **C**

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